Data Analytics Project 2 –
Exploratory Data Analysis (EDA) on E-Commerce Dataset

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Project Objective:

- Perform Exploratory Data Analysis (EDA) on an E-Commerce dataset. Understand patterns, trends, and distributions.
- Generate meaningful insights using SQL queries.

Tools Used:

- MySQL Workbench

Dataset Information:

- **Dataset:** E-Commerce Sales Dataset
- **Records:** 1000
- **Data Type**: Structured Data

Main Columns

- OrderID
- CustomerID
- Product
- Quantity
- UnitPrice
- TotalPrice
- PaymentMethod
- OrderStatus
- ReferralSource

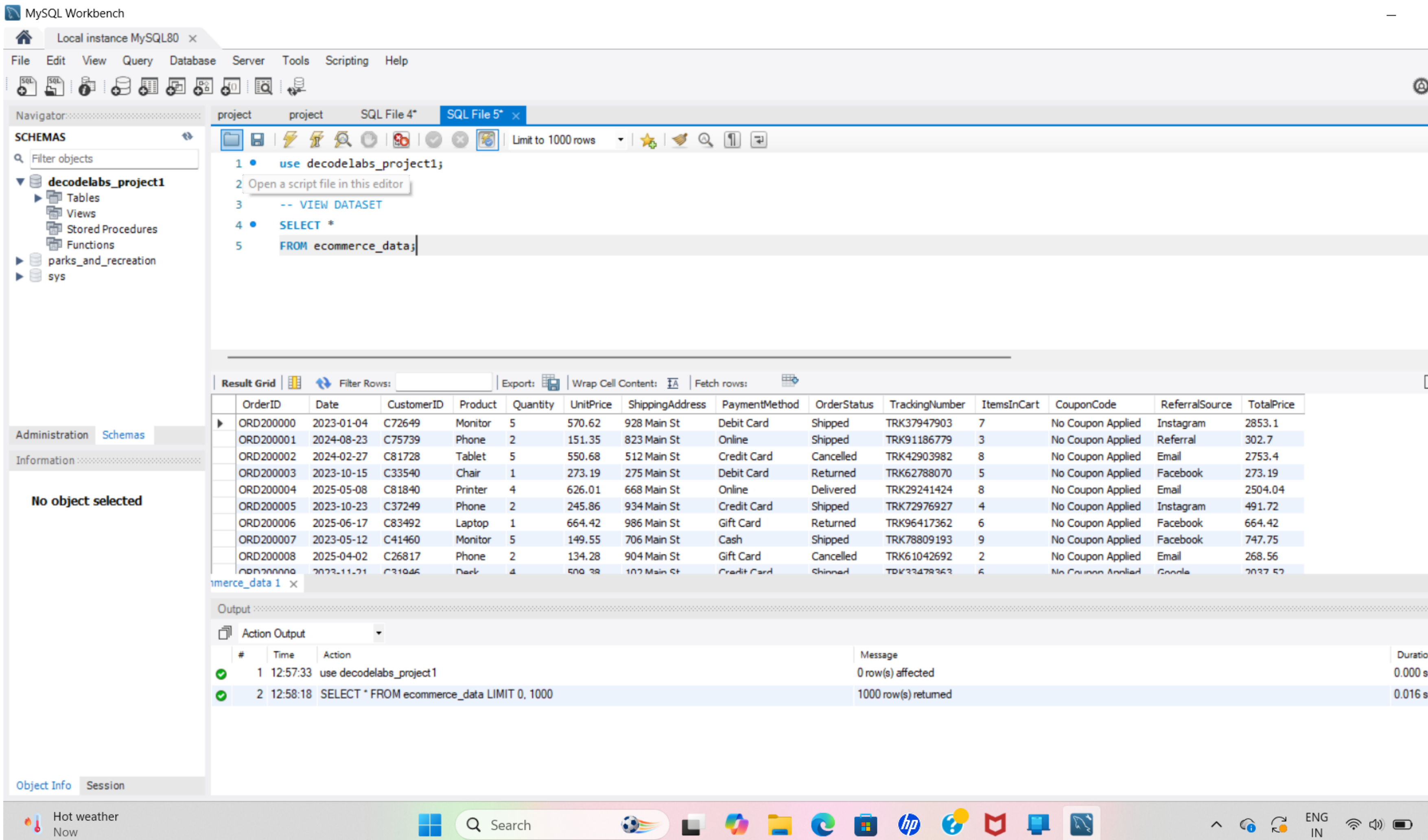

Fig 1: Dataset

Statistics:
Analysis Performed:

- Total Records
- Average Order Value
- Highest Order Value
- Lowest Order Value

Results:

- **Highest Order Value:** 3456.40

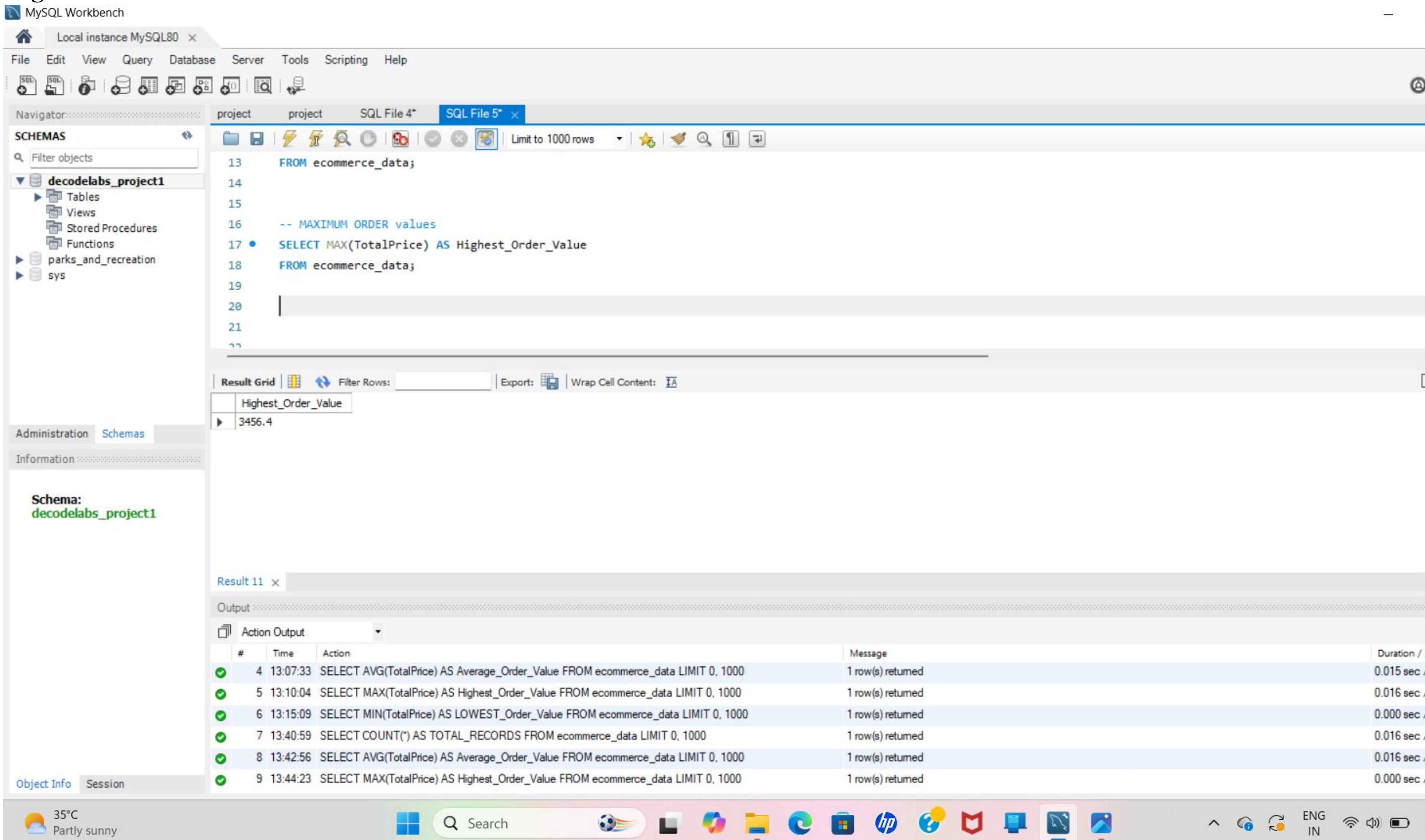

Fig 2 : Maximum Order Value

- **Lowest Order Value:** 11.39

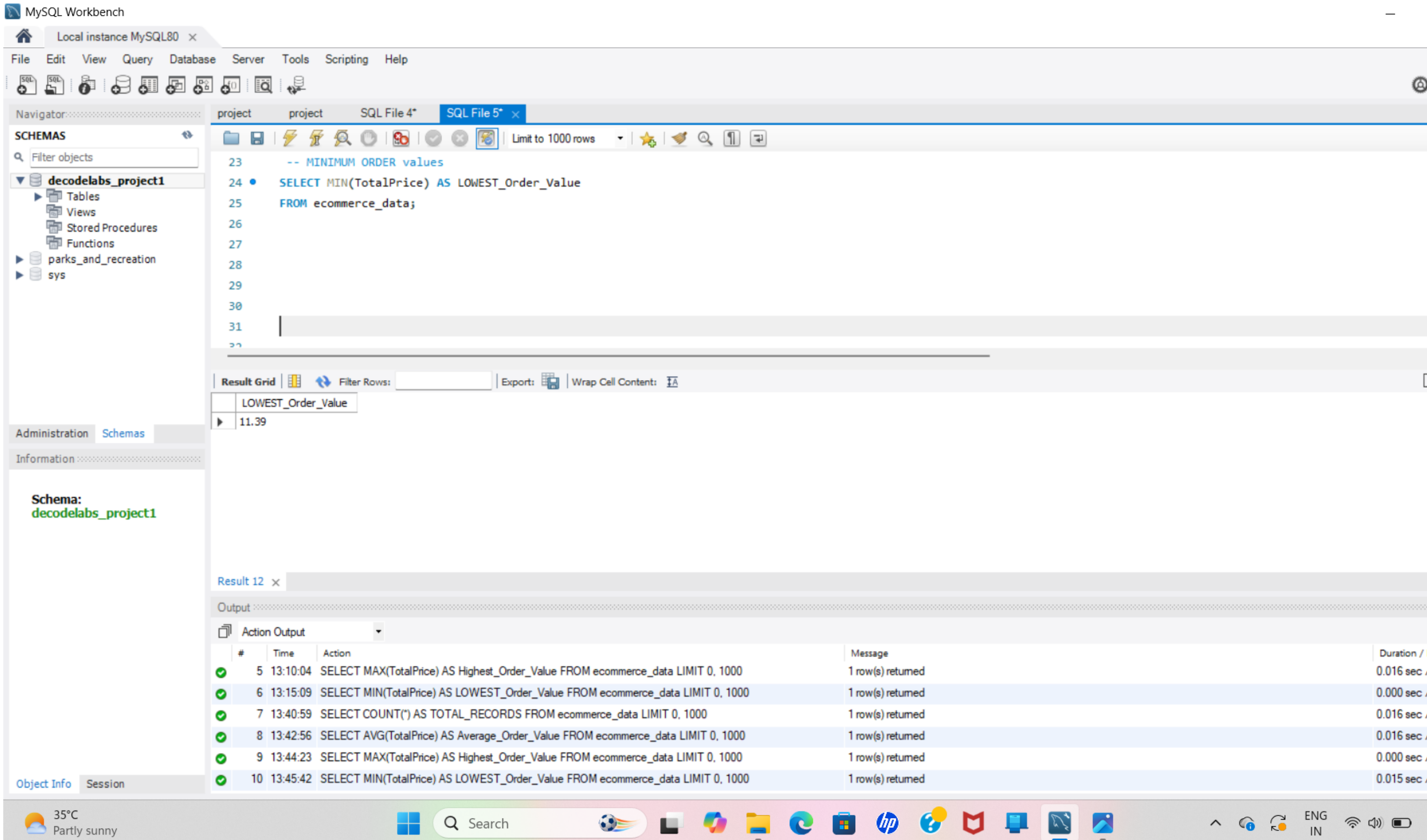

Fig 3 : Minimum Order Value

Product Analysis:
Queries Performed:

- Top Selling Products
- Revenue by Product

Observation:
- Instagram generated the highest customer traffic.

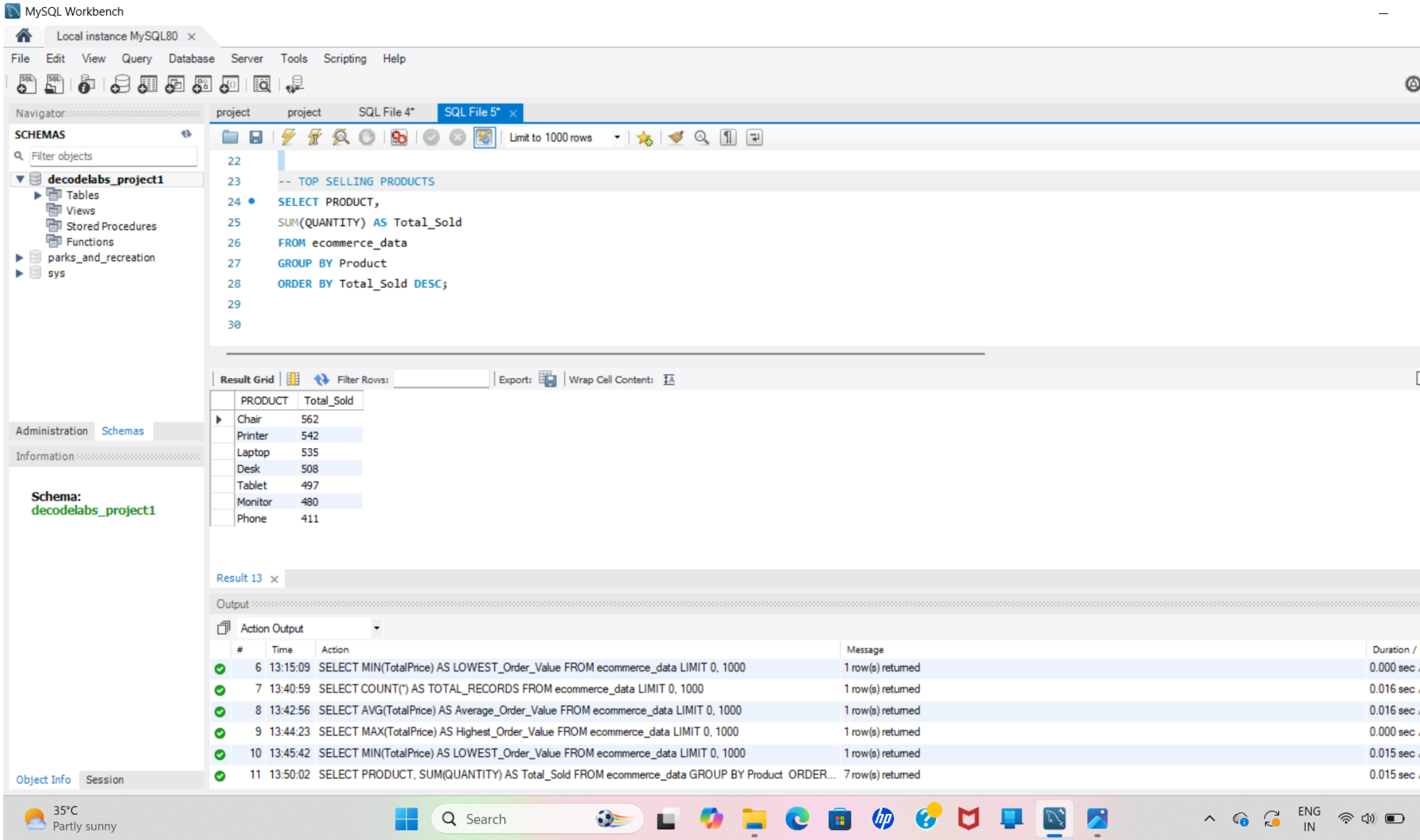

Fig 4 : Top Selling Products

Payment Analysis:

- Orders by Payment Method
- Online
- Cash
- Credit Card
- Debit Card
- Gift Card
- Observation
- Online payments recorded the highest number of orders.

Order Status Analysis:

- Order Status
- Shipped
- Delivered
- Cancelled
- Returned
- Pending
- Observation
- Cancelled orders were slightly higher than the other order statuses.

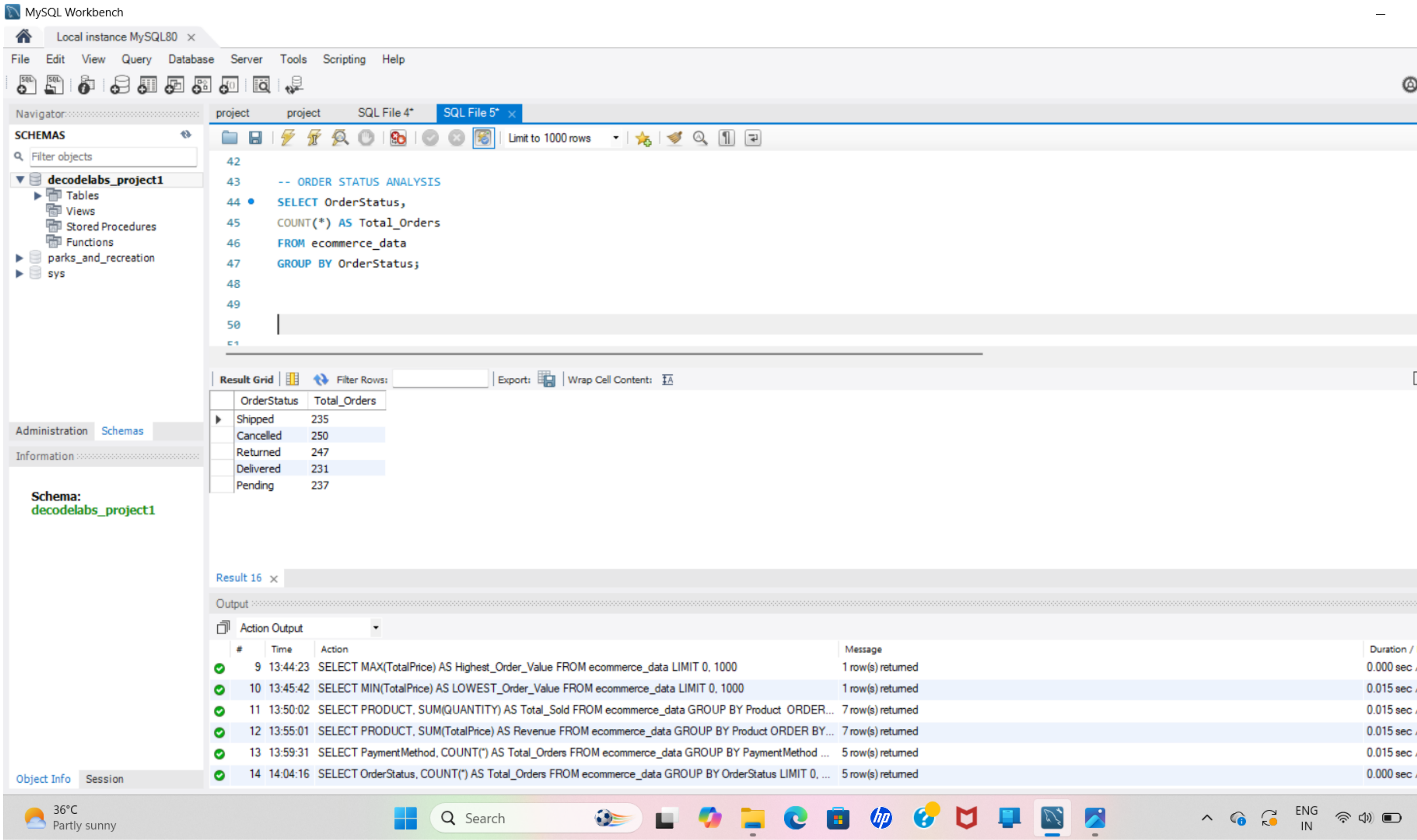

Fig 5: Payment Analysis

Referral Source Analysis:

- Referral Sources
- Instagram
- Email
- Google
- Facebook
- Referral
- Observation
- Instagram generated the highest customer traffic.

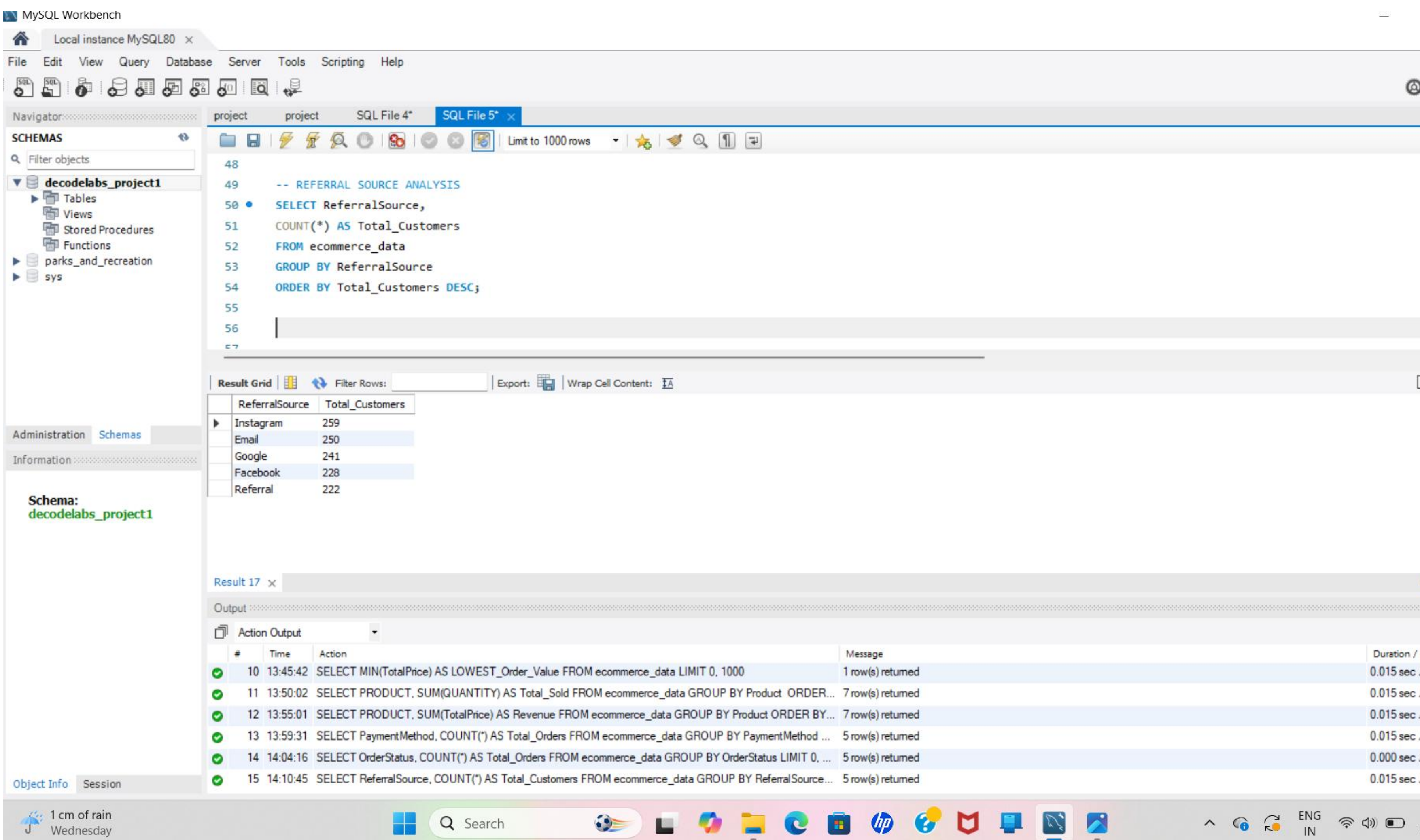

Fig 6 : Referral Source Analysis

Monthly Sales Trend:

- Sales Trend Analysis
- Monthly Revenue
- Sales Distribution
- Observation
- Sales varied across different months.

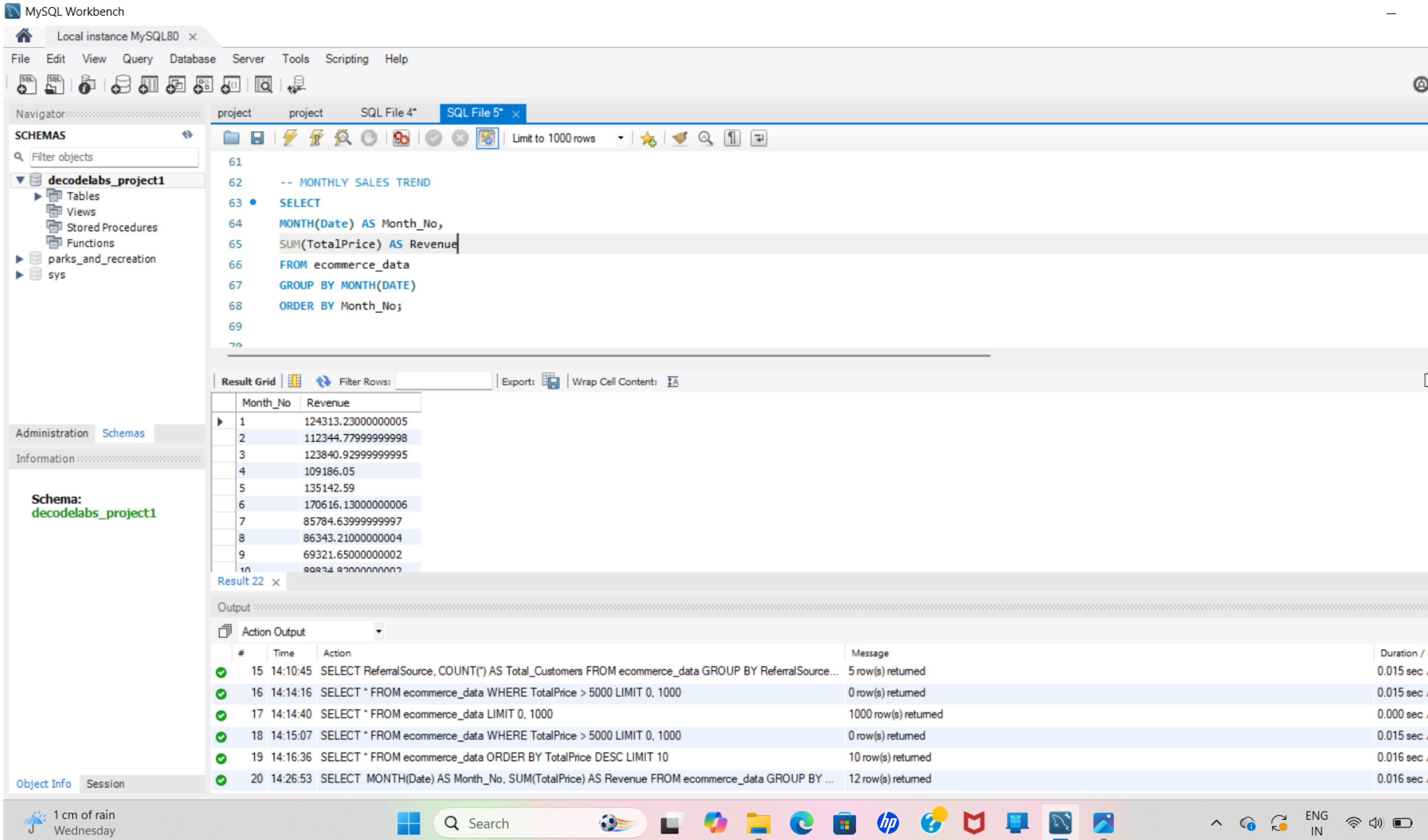

Fig 7 :Monthly Sales Trend

Outlier Detection:

- Outlier Analysis
- Checked for unusually high-value orders.
- Observation
- No significant outliers above the selected threshold were found.

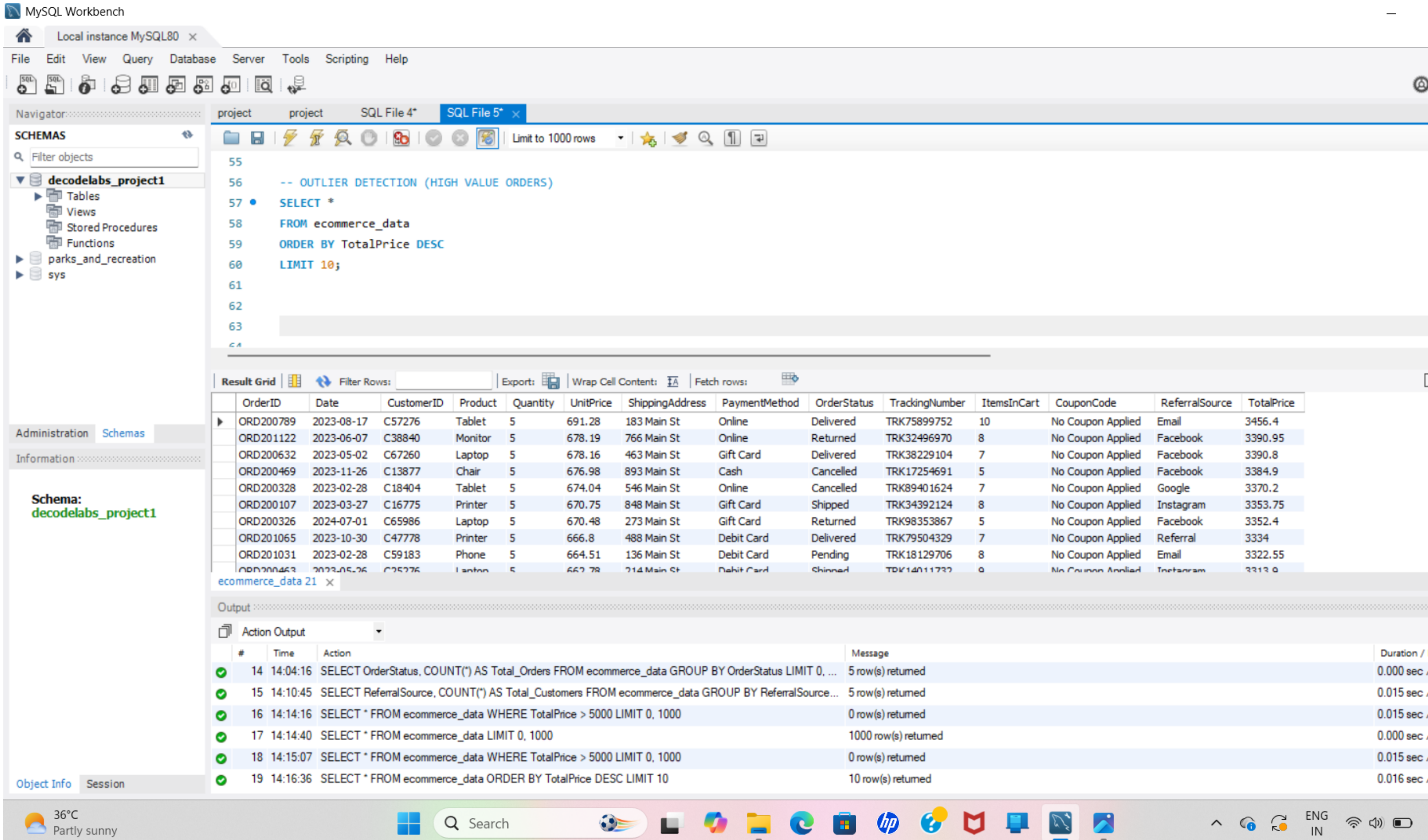

Fig 8 : Outlier Detection

Key Insights:

- Successfully performed Exploratory Data Analysis using SQL.
- Identified customer purchasing trends.
- Analyzed payment methods.
- Studied order status distribution.
- Examined referral source performance.
- Evaluated product revenue and sales trends.

Conclusion:

- Successfully completed Exploratory Data Analysis on the E-Commerce dataset.
- SQL queries helped uncover valuable business insights.
- The project strengthened practical skills in SQL, data analysis, and data interpretation.

THANK YOU